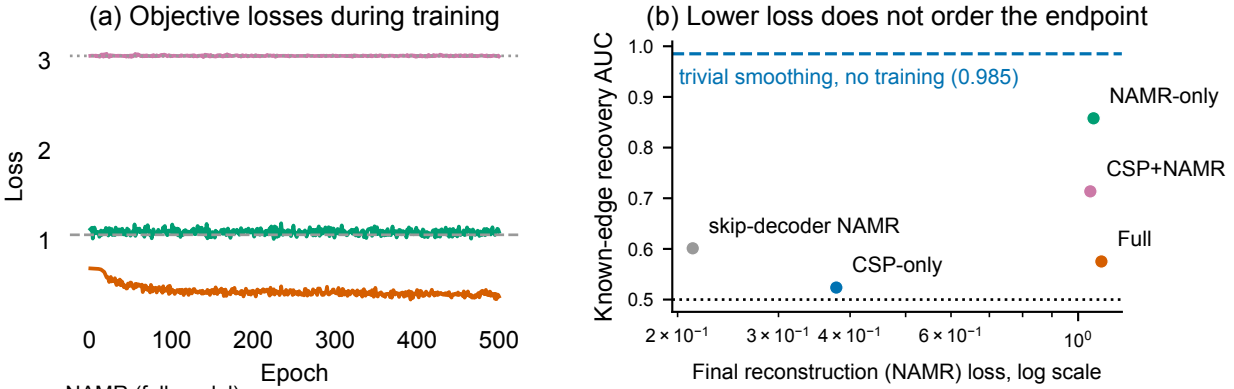
(a) Objective losses during training
Loss
3
2
1
0
100
200
300
400
500
Epoch
NAMR (full model)
CSP
C3L (InfoNCE)
ln N_{cancer} (C3L uniform-prototype level)
constant-mean reference (standardized NAMR target)
(b) Lower loss does not order the endpoint
Known-edge recovery AUC
1.0
0.9
0.8
0.7
0.6
0.5
trivial smoothing, no training (0.985)
NAMR-only
CSP+NAMR
Full
skip-decoder NAMR
CSP-only
2×10^{-1}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}
Final reconstruction (NAMR) loss, log scale